Practice workbook · Drawing the graph from your table

Plotting the points from a completed table on the printed grid and joining them with a smooth curve over the given interval.

15 question parts: 1 worked-example retry and 14 further practice parts.
Name: _____ Class: _____ Date: _____

Practise one question type at a time. Attempt the part inside the red boundary; the rest of the question and any figures are kept for context. Write your working in the answer space.

The first question is a retry of the website worked example, without its solution. Check your answers online with 'Check my answer', or ask your teacher for the answer booklet.

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Type ID: FPM-GRA-02

1. Question 7 · part b · 2 marks

2025 JUN · Paper 02

Worked-example retry · Attempt part b only.

7 (a) Complete the table of values for $y = \log_3(4-x)+3$ giving your answers to 2 decimal places.

x	0	0.5	1	1.5	2	2.5	3	3.5
y	4.26		4		3.63		3	2.37

(2)

(b) On the grid opposite, draw the graph of $y = \log_3(4-x)+3$ in the interval $0 \leqslant x \leqslant 3.5$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate,

to one decimal place, of the root of the equation $3^{2x-5} - (4-x)^3 = 0$ in the interval $0 \leqslant x \leqslant 3.5$

(6)

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1. Question 7 · part b · 2 marks · Working and answer

Working and answer

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2. Question 8 · part b · 2 marks

2019 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

8 (a) Complete the table of values for $y = 2 + \ln(2x + 1)$ giving your answers to 2 decimal places.

(2)

x	0	0.25	0.5	1	1.5	2	3
y	2			3.10	3.39	3.61	

(b) On the grid opposite, draw the graph of $y = 2 + \ln(2x + 1)$ for $0 \leqslant x \leqslant 3$

(2)

(c) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $\ln(2x + 1) = 3x - 4$ in the interval $0 \leqslant x \leqslant 3$

(3)

(d) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $e^{(6-x)} - (2x + 1)^2 = 0$ in the interval $0 \leqslant x \leqslant 3$

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P 6 1 8 8 2 A 0 2 0 3 6

2. Question 8 · part b · 2 marks · Working and answer

Working and answer

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3. Question 7 · part b · 2 marks

2023 JAN · Paper 01

Further practice · Attempt part b only.

7 (a) Complete the table of values for

$$y = 0.5^{\left(\frac{x}{3}+1\right)} + 2$$

giving each value to 2 decimal places where appropriate.

x	–6	–5	–4	–3	–2	–1	0
y	4	3.59	3.26				2.5

(2)

(b) On the grid opposite, draw the graph of $y = 0.5^{\left(\frac{x}{3}+1\right)} + 2$ for $-6 \leqslant x \leqslant 0$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$\log_2(2x+2)^3 + x + 3 = 0 \text{ in the interval } -6 \leqslant x \leqslant 0$$

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3. Question 7 · part b · 2 marks · Working and answer

Working and answer

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4. Question 4 · part b · 2 marks

2020 NOV · Paper 02 (regional R)

Further practice · Attempt part b only.

4 (a) Complete the table of values for $y = 2x + 1 + \frac{2}{x^2}$

Give your answers to 2 decimal places where appropriate.

x	0.5	1	1.5	2	2.5	3	3.5
y		5			6.32		8.16

(2)

(b) On the grid opposite, draw the graph of $y = 2x + 1 + \frac{2}{x^2}$ for $0.5 \leqslant x \leqslant 3.5$

(2)

(c) Use your graph to obtain estimates, to 1 decimal place, of the roots of the equation

$$2x + \frac{2}{x^2} = 7 \quad \text{in the interval } 0.5 \leqslant x \leqslant 3.5$$

(2)

(d) By drawing a suitable straight line on the grid, obtain estimates, to 1 decimal place, of the roots of the equation

$$\frac{3x}{2} + \frac{2}{x^2} = 5 \quad \text{in the interval } 0.5 \leqslant x \leqslant 3.5$$

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4. Question 4 · part b · 2 marks · Working and answer

Working and answer

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5. Question 8 · part b · 2 marks

2022 JAN · Paper 02

Further practice · Attempt part b only.

8 (a) Given that x is measured in radians, complete the table of values for

$$y = \sin x - 3\cos 2x - 0.5$$

Give your answers to one decimal place.

x	0	0.5	0.8	1	1.6	2	2.5	3
y			0.3	1.6			–0.8	–3.2

(2)

(b) On the grid opposite, draw the graph of $y = \sin x - 3\cos 2x - 0.5$ for $0 \leqslant x \leqslant 3$

(2)

(c) Use a formula from page 2 to show that $\cos 2A = 1 - 2\sin^2 A$

(2)

$$f(x) = 2\sin x + 12\sin^2 x - x - 5$$

(d) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, of the roots of the equation $f(x) = 0$ in the interval $0 \leqslant x \leqslant 3$

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5. Question 8 · part b · 2 marks · Working and answer

Working and answer

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6. Question 6 · part b · 2 marks

2020 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

6 (a) Complete the table of values for

$$y = x - \frac{3}{x^2}$$

giving your answers to one decimal place where appropriate.

x	0.5	1	1.5	2	3	4	5	6
y	–11.5			1.3	2.7			5.9

(2)

(b) On the grid opposite, draw the graph of $y = x - \frac{3}{x^2}$ for $0.5 \leqslant x \leqslant 6$

(2)

(c) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, of each of the two roots of the equation

$$2x^3 - 6x^2 + 3 = 0$$

in the interval $0.5 \leqslant x \leqslant 6$

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6. Question 6 · part b · 2 marks · Working and answer

Working and answer

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7. Question 5 · part b · 2 marks

2022 JUN · Paper 02 (regional R)

Further practice · Attempt part b only.

5 (a) Complete the table of values for $y = e^{3x-2}$ giving your answers to 2 decimal places.

x	0	0.25	0.5	0.75	1
y	0.14				2.72

(2)

(b) On the grid opposite, draw the graph of $y = e^{3x-2}$ for $0 \leqslant x \leqslant 1$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $3x = 2 + \ln(3 - x)$

(3)

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7. Question 5 · part b · 2 marks · Working and answer

Working and answer

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8. Question 8 · part b · 2 marks

2021 NOV · Paper 02

Further practice · Attempt part b only.

8 (a) Complete the table of values for $y = 2x + \frac{3}{x^2} - 3$ giving your answers to 2 decimal places where appropriate.

x	0.5	0.75	1	1.5	2	3	4	5
y	10		2				5.19	7.12

(2)

(b) On the grid opposite, draw the graph of $y = 2x + \frac{3}{x^2} - 3$ for $0.5 \leqslant x \leqslant 5$

(2)

(c) By drawing a suitable straight line on the grid, obtain estimates, to one decimal place, of the roots of the equation $4x^3 - 10x^2 + 3 = 0$ for $0.5 \leqslant x \leqslant 5$

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8. Question 8 · part b · 2 marks · Working and answer

Working and answer

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9. Question 4 · part b · 2 marks

2024 NOV · Paper 02

Further practice · Attempt part b only.

4 (a) Complete the table of values for $y = \log_{10}(6x-1) - x$ giving your answers to 2 decimal places.

x	0.25	0.5	1	1.5	2	2.5	3
y		–0.20	–0.30	–0.60			

(2)

(b) On the grid opposite, draw the graph of $y = \log_{10}(6x-1) - x$ for $0.25 \leqslant x \leqslant 3$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$10^{\frac{3x-4}{2}} = 6x - 1 \quad \text{in the interval } 0.25 \leqslant x \leqslant 3$$

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9. Question 4 · part b · 2 marks · Working and answer

Working and answer

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10. Question 9 · part b · 2 marks

2022 JAN · Paper 01 (regional R)

Further practice · Attempt part b only.

9 (a) Complete the table of values for

$$y = \frac{x^3 + 4}{5 - x}$$

giving your answers to 2 decimal places.

x	−2	−1.5	−1	−0.5	0	0.5	1	1.5	2
y	−0.57	0.10	0.5		0.8		1.25		4

(2)

(b) On the grid opposite, draw the graph of

$$y = \frac{x^3 + 4}{5 - x} \quad \text{for } -2 \leqslant x \leqslant 2$$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation $x^3 - x^2 + 8x - 11 = 0$ in the interval $-2 \leqslant x \leqslant 2$

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10. Question 9 · part b · 2 marks · Working and answer

Working and answer

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11. Question 7 · part b · 2 marks

2025 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

7 (a) Complete the table of values for $y = 3\log_3\left(x^2 - x\right) - 2x$ giving your answers to 2 decimal places.

x	1.3	1.5	2	2.5	3	3.5
y		−3.79	−2.11			

(2)

(b) On the grid opposite, draw the graph of $y = 3\log_3\left(x^2 - x\right) - 2x$ for $1.3 \leqslant x \leqslant 3.5$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$3^{\frac{4}{3}x} = 9\left(x^2 - x\right)^3 \quad \text{in the interval } 1.3 \leqslant x \leqslant 3.5$$

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11. Question 7 · part b · 2 marks · Working and answer

Working and answer

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12. Question 4 · part b · 2 marks

2023 JUN · Paper 01 (regional R)

Further practice · Attempt part b only.

4 (a) Complete the table of values for $y = \frac{x}{2} + 6e^{-2x} + 1$

giving your answers to one decimal place.

x	0	1	1.5	2	3	4	5	6
y	7		2.0			3.0		4.0

(2)

(b) On the grid opposite, draw the graph of $y = \frac{x}{2} + 6e^{-2x} + 1$ for $0 \leqslant x \leqslant 6$

(2)

(c) By drawing a suitable straight line on your graph, obtain estimates, to one decimal place, of the roots of the equation

$$2x + \ln(24 - 5x) = \ln 36$$

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12. Question 4 · part b · 2 marks · Working and answer

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13. Question 8 · part b · 2 marks

2025 NOV · Paper 01

Further practice · Attempt part b only.

8 (a) Complete the table of values for $y = 3^x - 2$ giving your answers to one decimal place.

x	-1	-0.5	0	0.25	0.5	0.75	1	1.5	2
y	-1.7		-1				1	3.2	7

(2)

(b) On the grid opposite, draw the graph of $y = 3^x - 2$ for $-1 \leqslant x \leqslant 2$

(2)

(c) Use your graph to obtain an estimate, to one decimal place, of the value of $\log_3 4.5$

Show clearly how you have used your graph.

(3)

(d) By drawing a straight line on your grid obtain an estimate, to one decimal place, of the root of the equation $\log_3(8-3x) - x = 0$ in the interval $-1 \leqslant x \leqslant 2$

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13. Question 8 · part b · 2 marks · Working and answer

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14. Question 7 · part b · 2 marks

2021 JUN · Paper 02

Further practice · Attempt part b only.

7 (a) Complete the table of values for $y = 3^{\frac{x}{4}} + 2$

Give your answers to 2 decimal places where appropriate.

(2)

x	0	1	2	3	4	5
y	3	3.32				5.95

(b) On the grid opposite, draw the graph of

$$y = 3^{\frac{x}{4}} + 2 \quad \text{for} \quad 0 \leqslant x \leqslant 5$$

(2)

(c) By drawing a suitable straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$\log_3(6 - 2x)^4 - x = 0$$

in the interval $0 \leqslant x \leqslant 5$

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14. Question 7 · part b · 2 marks · Working and answer

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15. Question 6 · part b · 2 marks

2022 JUN · Paper 01

Further practice · Attempt part b only.

6 (a) Complete the table of values for $y = 1 + 3e^{-x}$ giving your answers to 2 decimal places where appropriate.

(2)

x	0	0.25	0.5	1	1.5	2	3
y		3.34	2.82		1.67		1.15

(b) On the grid opposite, draw the graph of $y = 1 + 3e^{-x}$ for $0 \leqslant x \leqslant 3$

(2)

(c) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$x = e^{-x} \quad \text{in the interval } 0 \leqslant x \leqslant 3$$

(3)

(d) By drawing an appropriate straight line on the grid, obtain an estimate, to one decimal place, of the root of the equation

$$\ln(x-1)^3 = -3x \quad \text{in the interval } 0 \leqslant x \leqslant 3$$

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15. Question 6 · part b · 2 marks · Working and answer

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